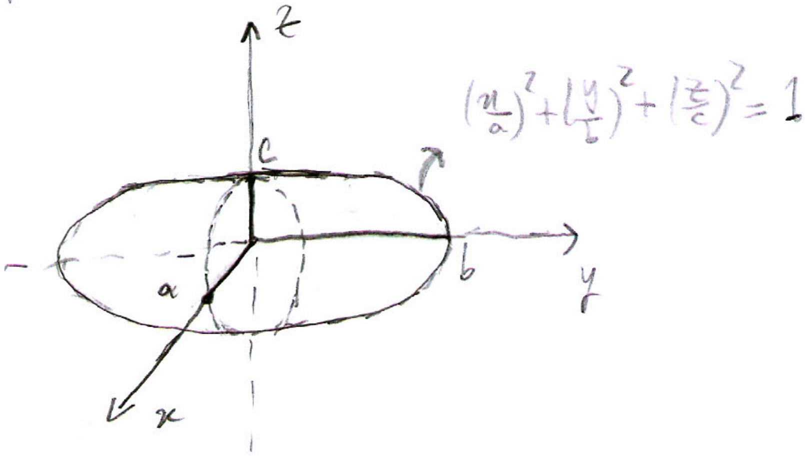
24ª aula prática

Exercícios resolvidos: 96b, 96c, 96d, 98.

96 b) Pretendemos calcular o volume do elipsóide $\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2+\left(\frac{z}{c}\right)^2=1$ representado na seguinte figura,



usando as coordenadas esféricas "adaptadas"

$$\begin{cases} x = a\rho \operatorname{sen}\varphi \cos\theta \\ y = b\rho \operatorname{sen}\varphi \operatorname{sen}\theta \\ z = c\rho\cos\varphi \end{cases}, \quad \theta\in[0,2\pi],\ \varphi\in[0,\pi],\ \rho\in\mathbb{R}_0^+.$$

Em coordenadas esféricas consideradas tem

$$\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2+\left(\frac{z}{c}\right)^2=1 \Leftrightarrow \rho^2\cos^2\theta \operatorname{sen}^2\varphi+\rho^2\operatorname{sen}^2\theta\operatorname{sen}^2\varphi+\rho^2\cos^2\varphi=1$$

$$\Leftrightarrow \rho^2\operatorname{sen}^2\varphi(\cos^2\theta+\operatorname{sen}^2\theta)+\rho^2\cos^2\varphi=1$$

$$\Leftrightarrow \rho^2(\operatorname{sen}^2\varphi+\cos^2\varphi)=1 \Leftrightarrow \rho=1$$

Seja S o sólido limitado pelo elipsóide. Então temos

$$S = \left\{ (x,y,z) \in \mathbb{R}^3 : \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 \leq 1 \right\} =$$

$$= \left\{ (\rho, \theta, \varphi) \in \mathbb{R}_0^+ \times [0, 2\pi] \times [0, \pi] : 0 \leq \theta \leq 2\pi \ \wedge \right.$$

$$\left. \wedge \ 0 \leq \varphi \leq \pi \ \wedge \ 0 \leq \rho \leq 1 \right\}.$$

O jacobiano desta mudança de variáveis é dado por

$$\begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \varphi} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \varphi} \end{vmatrix} = \begin{vmatrix} a \operatorname{sen}\varphi \cos\theta & -a\rho \operatorname{sen}\varphi \operatorname{sen}\theta & a\rho \cos\varphi \cos\theta \\ b \operatorname{sen}\varphi \operatorname{sen}\theta & b\rho \operatorname{sen}\varphi \cos\theta & b\rho \cos\varphi \operatorname{sen}\theta \\ c \cos\varphi & 0 & -c\rho \operatorname{sen}\varphi \end{vmatrix}$$

$$= \left| (a\rho \operatorname{sen}\varphi \operatorname{sen}\theta) \begin{vmatrix} b \operatorname{sen}\varphi \operatorname{sen}\theta & b\rho \cos\varphi \operatorname{sen}\theta \\ c\cos\varphi & -c\rho \operatorname{sen}\varphi \end{vmatrix} + \right.$$

Teorema Laplace (2ª coluna)

$$\left. b\rho \operatorname{sen}\varphi \cos\theta \begin{vmatrix} a \operatorname{sen}\varphi \cos\theta & a\rho \cos\varphi \cos\theta \\ c \cos\varphi & -c\rho \operatorname{sen}\varphi \end{vmatrix} \right| =$$

$$= \left| a\rho \operatorname{sen}\varphi \operatorname{sen}\theta \cdot \left(-bc\rho \boxed{\operatorname{sen}^2\varphi} \operatorname{sen}\theta - bc\rho \boxed{\cos^2\varphi} \operatorname{sen}\theta \right) \right.$$

$$\left. + \, b\rho \operatorname{sen}\varphi \cos\theta \left(-ac\rho \boxed{\operatorname{sen}^2\varphi} \cos\theta - ac\rho \boxed{\cos^2\varphi} \cos\theta \right) \right| =$$

$$= \left| a\rho \sin\varphi \sin\theta \left(-bc\rho\sin\theta\right) + b\rho\sin\varphi\cos\theta\left(-ac\rho\cos\theta\right)\right|$$

$$= \left|-abc\rho^2 \sin\varphi \left(\sin^2\theta + \cos^2\theta\right)\right| = \left|-abc\rho^2\sin\varphi\right| =$$

$$= abc\rho^2\sin\varphi.$$

Então o volume do sólido S é dado por

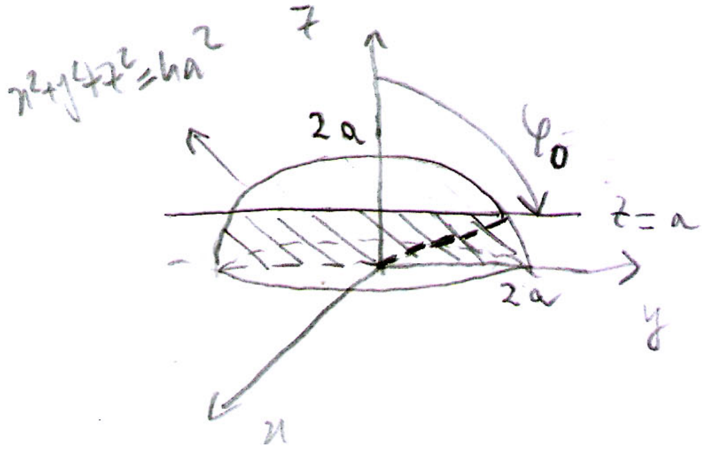
$$V = \iiint_S 1\, dx\, dy\, dz = \int_0^{2\pi}\int_0^{\pi}\int_0^{1} abc\rho^2 \sin\varphi \, d\rho\, d\varphi\, d\theta$$

$$= abc\int_0^{2\pi}\int_0^{\pi} \sin\varphi \left[\frac{\rho^3}{3}\right]_0^1 d\varphi\, d\theta = \frac{abc}{3}\int_0^{2\pi}\int_0^{\pi} \sin\varphi \, d\varphi\, d\theta$$

$$= \frac{abc}{3}\int_0^{2\pi}\left[-\cos\varphi\right]_0^{\pi} d\theta = \frac{abc}{3}\int_0^{2\pi} 2\, d\theta =$$

$$= \frac{abc}{3}\cdot 2\pi\cdot 2 = \frac{4\pi}{3}abc.$$

96 b) Calculemos o volume do sólido S limitado ~~superiormente~~ pela superfície de equação $x^2+y^2+z^2 = 4a^2$ e pelos planos $z=0$ e $z=a$, representado na seguinte figura:



Consideremos a mudança de variável para coordenadas esféricas

$$\begin{cases} x = \rho \sin\varphi \cos\theta \\ y = \rho \sin\varphi \sin\theta \\ z = \rho \cos\varphi \end{cases}, \theta \in [0, 2\pi], \varphi \in [0,\pi], \rho \in \mathbb{R}_0^+$$

cujo jacobiano sabemos já ser igual a $\rho^2 \sin\varphi$.

A equação do plano $z=a$ em coordenadas esféricas vem de

$$z = a \Leftrightarrow \rho\cos\varphi = a \Leftrightarrow \rho = \frac{a}{\cos\varphi}.$$

Relativamente ao ângulo φ do ponto de intersecção da superfície esférica com o plano $z=a$, notando que $x^2+y^2+z^2=4a^2 \Leftrightarrow \rho^2 = 4a^2 \Leftrightarrow \rho = 2a$. Temos então

$$\begin{cases} \rho = \frac{a}{\cos\varphi} \\ \rho = 2a \end{cases} \Leftrightarrow \begin{cases} \rho = 2a \\ \frac{a}{2a} = \cos\varphi \end{cases} \Leftrightarrow \begin{cases} \rho = 2a \\ \cos\varphi = \frac{1}{2} \end{cases} \Leftrightarrow \begin{cases} \rho = 2a \\ \varphi = \pi/3 . \end{cases}$$

Assim na figura acima temos $\varphi_0 = \pi/3$.

Assim

$$S=\left\{(\rho,\theta,\varphi):\left(0\le\varphi\le\frac{\pi}{3}\wedge 0\le\rho\le\frac{a}{\cos\varphi}\wedge 0\le\theta\le 2\pi\right)\vee\right.$$

$$\left.\vee\left(\frac{\pi}{3}\le\varphi\le\frac{\pi}{2}\wedge 0\le\rho\le 2a\wedge 0\le\theta\le 2\pi\right)\right\}.$$

Então

$$V=\iiint_S 1\,dx\,dy\,dz=\int_0^{2\pi}\int_0^{\frac{\pi}{3}}\int_0^{\frac{a}{\cos\varphi}}\rho^2\operatorname{sen}\varphi\,d\rho\,d\varphi\,d\theta$$

$$+\int_0^{2\pi}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\int_0^{2a}\rho^2\operatorname{sen}\varphi\,d\rho\,d\varphi\,d\theta=$$

$$=\int_0^{2\pi}\int_0^{\frac{\pi}{3}}\operatorname{sen}\varphi\left[\frac{\rho^3}{3}\right]_0^{a/\cos\varphi}d\varphi\,d\theta+\int_0^{2\pi}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\operatorname{sen}\varphi\left[\frac{\rho^3}{3}\right]_0^{2a}d\varphi\,d\theta$$

$$=\frac{1}{3}\int_0^{2\pi}\int_0^{\frac{\pi}{3}}\operatorname{sen}\varphi\,\frac{a^3}{\cos^3\varphi}\,d\varphi\,d\theta+\int_0^{2\pi}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\operatorname{sen}\varphi\cdot\frac{8a^3}{3}\,d\varphi\,d\theta=$$

$$=-\frac{a^3}{3}\int_0^{2\pi}\int_0^{\frac{\pi}{3}}(-\operatorname{sen}\varphi)\cos^{-3}\varphi\,d\varphi\,d\theta+\frac{8a^3}{3}\int_0^{2\pi}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\operatorname{sen}\varphi\,d\varphi\,d\theta=$$

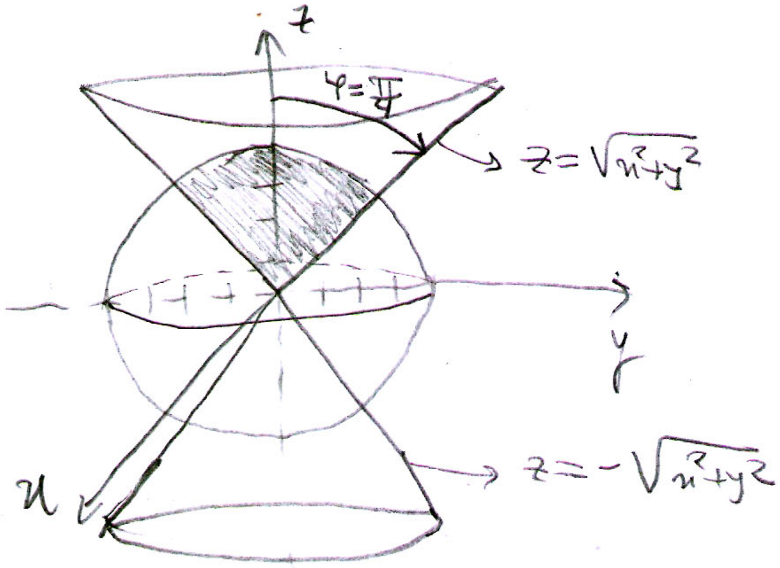
$$=-\frac{a^3}{3}\int_0^{2\pi}\left[\frac{\cos^{-2}\varphi}{-2}\right]_0^{\frac{\pi}{3}}d\theta+\frac{8a^3}{3}\int_0^{2\pi}\left[-\cos\varphi\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}d\theta=$$

$$=\frac{a^3}{6}\cdot\left(\frac{1}{\cos^2(\frac{\pi}{3})}-\frac{1}{\cos^2(0)}\right)\cdot 2\pi+\frac{8a^3}{3}\left(\underbrace{-\cos\tfrac{\pi}{2}}_{0}+\underbrace{\cos\tfrac{\pi}{3}}_{1/2}\right)\cdot 2\pi=$$

$$= \pi a^3 + \frac{8a^3}{3} \cdot \frac{1}{2} \cdot 2\pi = a^3\pi + \frac{8}{3}a^3\pi = \frac{11}{3}a^3\pi.$$

e) Comecemos por calcular o volume do sólido limitado superiormente pela superfície esférica $x^2+y^2+z^2=16$ e inferiormente pela superfície cónica $z^2=x^2+y^2$.

A representação geométrica do sólido pode ser feita da seguinte forma:



na qual o sólido está marcado a sombreado.

Usamos as coordenadas esféricas:

$$\begin{cases} x = \rho \operatorname{sen}\varphi \cos\theta \\ y = \rho \operatorname{sen}\varphi \operatorname{sen}\theta \\ z = \rho\cos\varphi \end{cases}, \theta \in [0,2\pi], \varphi \in [0,\pi], \rho \in \mathbb{R}_0^+$$

e $|J| = \rho^2 \operatorname{sen}\varphi$.

A superfície cónica em coordenadas esféricas é obtida por

$$z = \sqrt{x^2+y^2} \Leftrightarrow \rho\cos\varphi = \sqrt{\rho^2 \operatorname{sen}^2\varphi \cos^2\theta + \rho^2 \operatorname{sen}^2\varphi \operatorname{sen}^2\theta} \Leftrightarrow$$

$$\underset{\rho\neq 0}{\Leftrightarrow} \rho\cos\varphi = \rho \operatorname{sen}\varphi \Leftrightarrow$$

$$\underset{\cos\varphi\neq 0}{\Leftrightarrow} \operatorname{tg}\varphi = 1 \Leftrightarrow \varphi = \pi/4 .$$

Então

$$S = \{(\rho,\theta,\varphi) \in \mathbb{R}^+ \times [0,2\pi] \times [0,\pi] : 0 \leq \theta \leq 2\pi \wedge 0 \leq \rho \leq 4 \wedge$$
$$\wedge\ 0 \leq \varphi \leq \pi/4 \},$$

Então

$$V = \iiint_S 1 \, dx\,dy\,dz = \int_0^{2\pi}\int_0^4\int_0^{\pi/4} \rho^2 \operatorname{sen}\varphi \, d\varphi \, d\rho \, d\theta =$$

$$= \int_0^{2\pi}\int_0^4 \rho^2 \big[-\cos\varphi\big]_0^{\pi/4} \, d\rho \, d\theta =$$

$$= \int_0^{2\pi}\int_0^4 \rho^2 \left(-\frac{\sqrt{2}}{2}+1\right) d\rho \, d\theta =$$

$$= \left(1-\frac{\sqrt{2}}{2}\right)\int_0^{2\pi} \left[\frac{\rho^3}{3}\right]_0^4 d\theta = \left(1-\frac{\sqrt{2}}{2}\right)\cdot\frac{1}{3}\cdot 64 \cdot 2\pi =$$

$$= \frac{128}{3}\pi\left(1-\frac{\sqrt{2}}{2}\right)$$

7

Quanto às coordenadas do centro de massa temos

$$M = \iiint_S \underbrace{d(x,y,z)}_{K}\, dx\,dy\,dz =$$

$$= K \iiint_S 1\, dx\,dy\,dz = \frac{128\pi}{3}\left(1-\frac{\sqrt{2}}{2}\right) K.$$

Temos ainda

$$M_x = \iiint_S x\,K\, dx\,dy\,dz = \int_0^4 \int_0^{\pi/4} \int_0^{2\pi} K(\rho \operatorname{sen}\varphi \cos\theta) \overbrace{\rho^2 \operatorname{sen}\varphi}^{|J|}\, d\theta\, d\varphi\, d\rho$$

$$= K \int_0^4 \int_0^{\pi/4} \rho^3 \operatorname{sen}^2\varphi \int_0^{2\pi} \cos\theta\, d\theta\, d\varphi\, d\rho =$$

$$= K \int_0^4 \int_0^{\pi/4} \rho^3 \operatorname{sen}^2\varphi \underbrace{\left[\operatorname{sen}\theta\right]_0^{2\pi}}_{0} d\varphi\, d\rho = 0,$$

$$M_y = \iiint_S y\,K\, dx\,dy\,dz = \int_0^4 \int_0^{\pi/4} \int_0^{2\pi} K(\rho \operatorname{sen}\varphi \operatorname{sen}\theta) \overbrace{\rho^2 \operatorname{sen}\varphi}^{|J|}\, d\theta\, d\varphi\, d\rho$$

$$= K \int_0^4 \int_0^{\pi/4} \rho^3 \operatorname{sen}^2\varphi \int_0^{2\pi} \operatorname{sen}\theta\, d\theta\, d\varphi\, d\rho =$$

$$= K \int_0^4 \int_0^{\pi/4} \rho^3 \operatorname{sen}^2\varphi \underbrace{\left[-\cos\theta\right]_0^{2\pi}}_{0} d\varphi\, d\rho = 0.$$

Atendendo a que o sólido é homogéneo e à sua simetria em relação ao eixo dos ZZ os dois resultados anteriores são esperados, já que o centro de massa tem de se situar sobre o eixo dos ZZ.

$$M_z = \iiint_S z\, K \, dx\, dy\, dz =$$

$$= K \int_0^4 \int_0^{\pi/4} \int_0^{2\pi} (\rho \cos\varphi)(\rho^2 \operatorname{sen}\varphi)\, d\theta\, d\varphi\, d\rho =$$

$$= 2\pi K \int_0^4 \rho^3 \int_0^{\pi/4} \cos\varphi \operatorname{sen}\varphi \, d\varphi\, d\rho =$$

$$= 2\pi K \int_0^4 \rho^3 \left[\frac{\operatorname{sen}^2\varphi}{2}\right]_0^{\pi/4} d\rho =$$

$$\boxed{\int u^\alpha u' = \frac{u^{\alpha+1}}{\alpha+1} \quad (\alpha \neq -1)}$$

$$= \frac{2\pi K}{2} \int_0^4 \rho^3 \cdot \left(\frac{\sqrt{2}}{2}\right)^2 d\rho =$$

$$= \frac{\pi K}{2} \left[\frac{\rho^4}{4}\right]_0^4 = \frac{\pi K}{2} \cdot \frac{1}{4} \cdot 4^4 = 32\pi K.$$

Então o centro de massa é

$$CM = \left(\frac{M_x}{M}, \frac{M_y}{M}, \frac{M_z}{M}\right) = \left(0, 0, \frac{32\pi K}{\frac{128}{3}\pi\left(1-\frac{\sqrt{2}}{2}\right)K}\right)$$

$$= \left(0, 0, \frac{32.3}{128\left(1-\frac{\sqrt{2}}{2}\right)}\right) = \left(0, 0, \frac{3}{2(2-\sqrt{2})}\right).$$

98. Seja $f: \mathbb{R}^3 \to \mathbb{R}$ contínua em

$B(c,r) = \{(x,y,z) \in \mathbb{R}^3 : (x-c_1)^2 + (y-c_2)^2 + (z-c_3)^2 \leq r^2\}$,

i.e. (x,y,z) um ponto da esfera de centro $c = (c_1, c_2, c_3)$ e raio r.

Então sabemos que f é integrável em $B(c,r)$ e como $B(c,r)$ é um conjunto compacto sabemos pelo Teorema Weierstrass que f tem máximo M e mínimo m em $B(c,r)$ com $m, M \in \mathbb{R}$.

Assim temos

$$\underbrace{f(x_0,y_0,z_0)}_{m} \leq f(x,y,z) \leq \underbrace{f(x_1,y_1,z_1)}_{M} \ \forall (x,y,z) \in B(c,r)$$

para certos $(x_0,y_0,z_0), (x_1,y_1,z_1) \in B(c,r)$.

Então vem que

$$\iiint_{B(c,r)} f(x_0,y_0,z_0)\,dx\,dy\,dz \leq \iiint_{B(c,r)} f(x,y,z)\,dx\,dy\,dz \leq \iiint_{B(c,r)} f(x_1,y_1,z_1)\,dx\,dy\,dz$$

ou seja

$$f(x_0,y_0,z_0) \iiint_{B(c,r)} 1\,dx\,dy\,dz \leq \iiint_{B(c,r)} f(x,y,z)\,dx\,dy\,dz \leq f(x_1,y_1,z_1) \iiint_{B(c,r)} 1\,dx\,dy\,dz$$

Seja $V = \iiint_{B(c,r)} 1 \, dx\,dy\,dz$, o volume de $B(c,r)$.

Dividindo os membros das desigualdades anteriores por $V>0$ vem

$$f(x_0,y_0,z_0) \leq \overbrace{\boxed{\frac{\iiint_{B(c,r)} f(x,y,z)\,dx\,dy\,dz}{V}}}^{K} \leq f(x_1,y_1,z_1).$$

Então mais uma vez por f ser contínua em $B(c,r)$ o teorema de Bolzano garante que existe um ponto $(\bar{x},\bar{y},\bar{z}) \in \mathbb{R}^3$ no segmento que une (x_0,y_0,z_0) a (x_1,y_1,z_1) tal que

$$K = \frac{1}{V} \iiint_{B(c,r)} f(x,y,z)\,dx\,dy\,dz = f(\bar{x},\bar{y},\bar{z})$$

e portanto existe $(\bar{x},\bar{y},\bar{z}) \in B(c,r)$ tal que

$$\iiint_{B(c,r)} f(x,y,z)\,dx\,dy\,dz = f(\bar{x},\bar{y},\bar{z}).V$$